PROJECT 09: Multi-Currency Travel Budget Analysis

1. Problem Statement

Convert travel expenses into a home currency using supplied exchange rates, include conversion fees, compare spending with a budget and identify the largest category.

2. Objectives

• Convert expenses from multiple currencies.
• Add a percentage conversion fee.
• Calculate total spending and budget balance.
• Identify the largest converted expense category.

3. Assumptions

• Exchange rates are illustrative home-currency units per one foreign-currency unit.
• A 2% conversion fee is applied to each expense after conversion.
• Every expense currency must have a rate in the supplied rate table.
• The budget is stated in the home currency (rupees).

4. Algorithm

1. Read the exchange-rate table, expenses and budget.
2. Check that each expense currency has a rate and amount is non-negative.
3. Convert each expense using its exchange rate.
4. Apply the conversion fee.
5. Sum converted expenses and compare with budget.
6. Identify and print the largest category.

5. Flowchart

6. Pseudocode

```
START
  total = 0
  FOR each expense
    IF currency rate missing: REPORT invalid currency
    ELSE
      converted = amount * exchange_rate
      with_fee = converted * (1 + fee_rate)
      SAVE category and with_fee
      total = total + with_fee
  remaining = budget - total
  largest = category with maximum with_fee
  DISPLAY expenses, total, remaining, largest
END
```

7. Python Code

```
rates = {"USD": 83.0, "EUR": 90.0, "GBP": 105.0}
expenses = [
    {"category": "Hotel", "amount": 200, "currency": "USD"},
    {"category": "Food", "amount": 100, "currency": "EUR"},
    {"category": "Transport", "amount": 60, "currency": "GBP"},
]
```

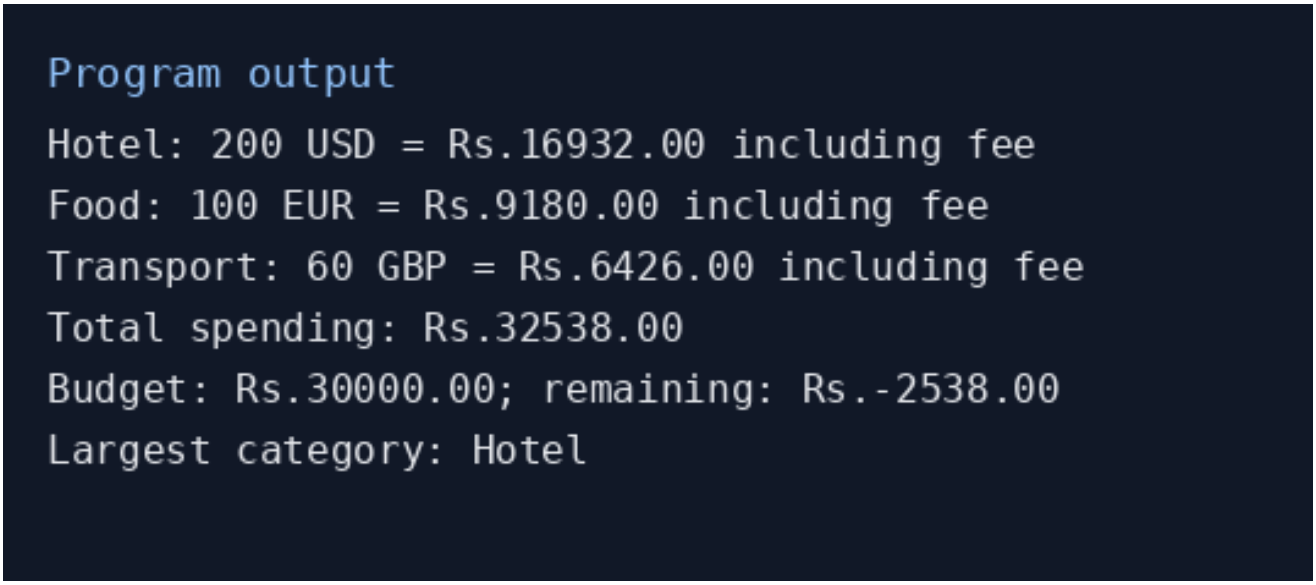
```
budget, fee_rate = 30000, 0.02
converted = []
for e in expenses:
    if e["amount"] < 0 or e["currency"] not in rates:
        print(f"{e['category']}: Invalid amount or currency")
        continue
    total = e["amount"] * rates[e["currency"]] * (1 + fee_rate)
    converted.append((e["category"], total))
    print(f"{e['category']}: {e['amount']} {e['currency']} = Rs.{total:.2f} including fee")
grand = sum(v for _, v in converted)
print(f"Total spending: Rs.{grand:.2f}")
print(f"Budget: Rs.{budget:.2f}; remaining: Rs.{budget-grand:.2f}")
if converted:
    print(f"Largest category: {max(converted, key=lambda x: x[1])[0]}")
```

8. Test Cases

Test	Input / condition	Expected result
USD conversion	USD 200 at rate 83 with 2% fee	Rs.16,932
EUR conversion	EUR 100 at rate 90 with 2% fee	Rs.9,180
Unknown currency	Currency JPY without rate	Expense is rejected with invalid currency message
Budget comparison	Total below budget	Remaining budget is positive

9. Output Screenshot

The following screenshot-style image was generated from the captured output of the program execution.



Captured output

```
Hotel: 200 USD = Rs.16932.00 including fee
Food: 100 EUR = Rs.9180.00 including fee
Transport: 60 GBP = Rs.6426.00 including fee
Total spending: Rs.32538.00
Budget: Rs.30000.00; remaining: Rs.-2538.00
Largest category: Hotel
```

10. Conclusion

The program consolidates expenses in one currency, accounts for a conversion fee and makes budget overruns and the largest cost category visible.

Academic note: The program was executed with the sample data shown in the source code. Policies, rates, thresholds and exchange rates are illustrative and should be adjusted to the assignment's official requirements.